Hong Nguyen

Curriculum Vitae

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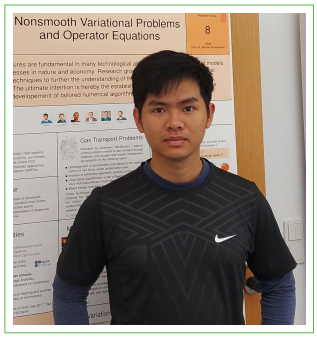
ResearchGate/Hong Quan Ba Nguyen



"The essence of mathematics lies in its freedom." - Georg Cantor
"It is true that a mathematician who is not somewhat of a poet, will never be a perfect mathematician." - Karl Weierstrass

Education

2020–now **PhD student enrolled in Humboldt-Universität zu Berlin**, *Weierstrass Institute for Applied Analysis and Stochastics* (WIAS), Berlin, Germany.
In the Research Group Nonsmooth Variational Problems & Operator Equations, Head: Prof. Michael Hintermüller, within the European Innovative Training Network Reduced Order Modelling, Simulation & Optimization of Coupled Systems (ROMSOC), in the Project 11 Optimal Shape Design of Air Ducts in Combustion Engines.

2018–2019 **Master de Sciences, Technologie, Santé, Mathematiques et Applications**, *Université de Rennes 1*, Rennes, France, *assez bien*.
Master 2 in Fundamental Mathematics and Applications (skipped Master 1).

2014–2018 **Bachelor of Science in Mathematics**, *University of Science* (HCMUS), *Viet Nam National University, Ho Chi Minh City* (VNU HCM), Vietnam, *very good*.
Specialized in Mathematical Analysis and Numerical Analysis.

2011–2014 **Elementary Mathematics**, *Ben Tre High School for Gifted Students* (CBT), Vietnam, *very good*.

Master Thesis

Title *Staggered and well-balanced discretization of shallow-water equations* [pdf][slide]

Supervisor Prof. Nicolas Seguin

Description This thesis investigates some staggered well-balanced finite volume schemes for the shallow water equations. A new staggered scheme, called "offset equilibrium", is developed to conquer the *dry-wet transition* issues where almost all the traditional finite volume schemes collapse. Higher-order extensions for this staggered scheme are also mentioned.

Research Interests

- **Mathematical Analysis** ▷ Partial Differential Equations (PDEs).
- **Numerical Analysis** ▷ Finite Difference Methods (FDMs), Finite Element Methods (FEMs), Finite Volume Methods (FVMs).
- **Mathematical Optimization** ▷ Shape Optimization, Topology Optimization.
- **Fluid Dynamics** ▷ Computational Fluid Dynamics (CFD), Turbulence.

Experience

Vocational

2019 **Master Internship**, IRMAR, Rennes, Bretagne, France.

Miscellaneous

2015–2017 **Olympiad Trainer**, *Ben Tre High School for Gifted Students* (CBT), Vietnam.
Trained the selected team of CBT for Vietnamese Mathematical Olympiads (VMOs).

Publications

[1] Nguyen Anh Dao, Jesús Ildefonso Díaz, Quan Ba Hong Nguyen. "Pointwise Gradient Estimates in Multi-dimensional Slow Diffusion Equations with a Singular Quenching Term". *Adv. Nonlinear Stud.* (2020). DOI https://doi.org/10.1515/ans-2020-2076.

Awards & Scholarships

Mar 2020–Aug 2021	Marie Skłodowska-Curie Actions (MSCA) PhD Fellowship, Innovative Training Network - European Industrial Doctorate Programs (MSCA-ITN-EID), Horizon 2020.
2018–2019	Lebesgue Master Scholarship for Master 2 program.
2017	Scholarship of Vietnam Institute for Advanced Studies in Mathematics (VIASM).
2016	Special Prize = First Prize in Algebra + First Prize in Analysis, in Vietnamese Mathematical Olympiad for College Students (VMC 2016).
2015	Second Prize in Algebra, Second Prize in Analysis, in Vietnamese Mathematical Olympiad for College Students (VMC 2015).
2012	Gold Medal #2, in Vietnamese Apr-30 Mathematical Olympiad for Central & Southern Vietnam.

Computer Skills

GitHub	GitHub/NQBH.
Basic	R, Fireshape, Firedrake, FEniCS, Python 3, STAR-CCM+.
Intermediate	UNIX/Linux [openSUSE, (K)Ubuntu], git, C/C++, OpenFOAM.
Advanced	TEX/LATEX, MATLAB.

Languages

Vietnamese	Mother tongue
English	Intermediate

Referees

[5]	**Prof. Michael Hintermüller**	*Primary PhD Supervisor*
[4]	**Prof. Nicolas Seguin**	*Master Supervisor*
[3]	**Dr. Nguyen Anh Dao**	*Bachelor Supervisor*
[2]	**Dr. Tan Trung Nguyen**	*Bachelor Teacher*
[1]	**Dr. Manh Khang Dao**	*Bachelor Teacher*

`Updated to Feb 3, 2021.`